From Manifolds to Number Theory: A Journey in Experimental Mathematics toward the Collatz Conjecture

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Abstract

We describe the development of a formalized approach to the Collatz conjecture that combines computational exploration, symbolic dynamics, Diophantine approximation, and machine-checked proof in Lean 4. Beginning with the observation that Collatz trajectories exhibit structure reminiscent of geodesics on a discrete manifold, we trace how experimental mathematics—large-scale computation over 10^{11} starting values, GPU-accelerated branch locus enumeration, and interactive theorem proving—led to a decomposition of the conjecture into a single irreducible statement: *bounded deficit*, formally equivalent to the conjecture itself. The formalization comprises approximately 7,800 lines across 34 files, resting on four literature axioms (Baker 1966, Rhin 1987, Hercher 2023, Weyl 1916) and eight explicit `sorry` declarations. This paper is intended as a companion to the formal proof, providing motivation, intuition, and the computational evidence that guided the formalization.

Contents

1 Introduction: why the Collatz conjecture resists proof

The Collatz conjecture asserts that iterating the map

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ 3n+1 & \text{if } n \equiv 1 \pmod 2, \end{cases} \tag{1}$$

from any positive integer n eventually reaches 1. Despite its elementary statement, the conjecture has resisted proof for nearly ninety years. Erdős famously remarked that "mathematics is not yet ready for such problems."

The difficulty is not a lack of evidence—every starting value up to 2^{68} has been verified computationally [4]—but a lack of *structure*. The map (1) alternates between two fundamentally different operations: division by 2 (contraction in the Archimedean metric) and multiplication by 3 followed by addition of 1 (expansion in the 3-adic metric). The interplay between these two metrics is what makes the problem hard.

This paper tells the story of how we arrived at a decomposition that makes the interplay explicit.

2 The manifold intuition

The project began with a simple observation: if we record the *parity sequence* of a Collatz trajectory—writing 0 for an even step and 1 for an odd step—the resulting binary string exhibits regularities reminiscent of geodesics on a Riemannian manifold.

2.1 The golden mean shift

The first structural constraint is that no two consecutive 1's appear in the parity sequence. This is because an odd step $n \mapsto 3n+1$ always produces an even number, so the next step is necessarily even. The set of all binary sequences with no consecutive 1's is the *golden mean shift* Σ_φ, a classical object in symbolic dynamics with topological entropy $\log \varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

This places every Collatz parity sequence inside a well-understood subshift of finite type (SFT). The equilibrium odd-step probability in Σ_φ is $p_{\text{eq}} = 1/\varphi^2 \approx 0.382$, which matches the empirically observed odd-step frequency with remarkable precision.

2.2 The torus and the branch locus

Given a parity sequence $\sigma \in \Sigma_\varphi$ of length t, the trajectory's position is determined by:

$$x_t \cdot 2^{\nu_2} = x_0 \cdot 3^{\nu_3} + \text{correction}, \tag{2}$$

where ν_2 and ν_3 count the even and odd steps, respectively. Taking this identity modulo 3^k for various k, the pair $(\nu_2 \bmod 3^k, \nu_3 \bmod 3^k)$ traces a path on the torus $(\mathbb{Z}/3^k\mathbb{Z})^2$.

The *branch locus* is the set of cells on this torus where the parity of the next step changes. Cells where $v_2 = 1$ (the 2-adic valuation of $3x+1$ is exactly 1) are *dangerous*—they contribute positively to the deficit $3\nu_3 - t$. Cells where $v_2 \geq 2$ are *safe*—they absorb the deficit from dangerous steps.

This geometric picture—trajectories as paths on a torus, with dangerous and safe regions—is what suggested a manifold-theoretic approach.

3 The computational laboratory

3.1 The branch locus program

To test the geometric intuition, we wrote a C program (`branch_locus.c`) that computes, for every starting value $n \leq N$:

- The complete parity sequence until convergence to 1.
- The branch locus cell counts on $(\mathbb{Z}/3^k\mathbb{Z})^2$ for $k = 1, 3, 9, 27, 81, 243, 729, 2187, 6561, 19683$.
- Transition counts between cell types.
- Shadow cells (pure-even cells that never host odd values).

The program uses OpenMP parallelization (24 threads) with a hybrid private/atomic grid strategy: small grids ($k \leq 162$) are thread-private (fitting in L2 cache), while large grids use shared atomic counters. Binary checkpoint/resume allows recovery from power interruptions.

3.2 The 10-billion run

A complete run over $N = 10^{10}$ (19.5 hours, 2.27×10^{12} total steps) confirmed:

- Mean odd-step proportion: $p_{\text{odd}} = 0.3324$.
- Zero "11" violations in 2.27×10^{12} transitions.
- Transition fractions: ee = 33.45%, eo = 33.16%, oe = 33.39%.

3.3 The 100-billion run (in progress)

At the time of writing, a run over $N = 10^{11}$ is approximately 63% complete (38 hours elapsed, ∼23 hours remaining). The odd-step proportion remains stable at $p_{\text{odd}} = 0.332$, consistent with the 10-billion data. This run will provide branch locus statistics at scales up to $k = 19683$ (5.8 GB grid), probing the large-scale structure of the Diophantine repeller.

3.4 GPU acceleration

To extend the branch locus computation, we implemented an OpenCL kernel (`gpu_branch_kernel.cl`) running on an NVIDIA RTX 5090 Laptop GPU. Pure convergence verification achieves 1,768M numbers/second (167× single-thread CPU). Branch counting with $k \leq 27$ (no global atomics) achieves 70M numbers/second. The GPU computation verified exact agreement with the CPU reference at $N = 10^7$.

3.5 The v_2 danger program

A separate program (`v2_danger.c`) focused on the 2-adic valuation structure:

- **Hensel attrition**: d consecutive $v_2 = 1$ steps require $x \equiv -1 \pmod{2^{d+1}}$, giving exact survival rate 2^{-d}.
- **Maximum run lengths**: $6 \to 9 \to 12$ for $N = 10^6 \to 10^7 \to 10^{10}$, growing as $\sim \log N$.
- **Hopping probability**: $P(D|D) \approx 0.06$, stable across scales. The absolute escape rate from danger is ∼94% per step.

These observations directly motivated the Diophantine repeller framework in the formalization.

4 The formalization: from intuition to Lean 4

4.1 Why formalize?

The Collatz conjecture is a problem where intuition is unreliable. Many plausible-sounding arguments fail on close inspection. Machine-checked proof in Lean 4 serves two purposes:

1. **Correctness**: Every step is verified by the Lean kernel, independent of human error or LLM hallucination.
2. **Precision**: The formalization forces us to make every assumption explicit. The "sorry" declarations pinpoint exactly where the mathematics gets hard.

The formalization comprises approximately 7,800 lines across 34 files, building in 3,109 Lean jobs.

4.2 The sorry inventory

The formalization rests on four external axioms citing established results:

A1. `baker_two_three` — Baker's theorem on linear forms in logarithms of 2 and 3 [1].

A2. `rhin_irrationality_measure` — Rhin's effective irrationality measure for $\log 2/\log 3$ [5].

A3. `hercher_no_small_cycle` — Hercher's computational verification that no non-trivial cycle has $\Delta_3 < 80$ [3].

A4. `weyl_equidistribution_of_irrational_rotation` — Weyl's equidistribution theorem for irrational rotations [8].

Beyond these axioms, eight `sorry` declarations remain:

- **Critical path** (2): `nu3_linear_bound` (`Drift.lean`) and `finite_deficit_bound` (`DiophantineRepeller.lean`). Both are formally equivalent to the Collatz conjecture.
- **Weyl bridge** (2): `equidistribution_implies_deficit_bounded` and `nu3_linear_bound_from_w` (`WeylEquidistribution.lean`).
- **Cycle analysis** (1): `steiner_cycle_large` (`SteinerCycle.lean`), handling cycles with $\Delta_3 \geq 80$.
- **Growth estimates** (3): `blaschke_factor_le_ratio`, `rising_factorial_le_pow`, and `poisson_jensen_blaschke` (`GrowthEstimates.lean`).

The critical path is:

$$\texttt{finite\_deficit\_bound} \xrightarrow{\text{proved}} \texttt{k\_bound\_of\_deficit\_bounded} \xrightarrow{\text{proved}} \text{K-bound} \xrightarrow{\text{proved}} \texttt{collatz\_conjectu}$$

Everything except `finite_deficit_bound` is machine-checked.

4.3 The reduction to bounded deficit

The formalization reduces the Collatz conjecture to a single statement about the deficit function:

Theorem 4.1 (Bounded deficit $\Leftrightarrow$ Collatz)**.** *For every* $n \geq 1$*,* $\mathrm{collatzReaches}(n)$ *if and only if there exists* $D \geq 0$ *such that for all* t*:*

$$\delta(n, t) \leq D.$$

That is: the Collatz conjecture is equivalent to the assertion that the deficit $\delta(t) = 3\nu_3(t) - t$ is bounded along every trajectory. The deficit budget framework (`DeficitBudget.lean` and `DiophantineRepeller.lean`) provides a sorry-free reduction from bounded deficit to the K-bound, and thence to the Collatz conjecture. The equivalence `nu3_linear_bound` $\leftrightarrow$ `collatzReaches` is proved in `Conclusion.lean`.

5 The conflict of metrics

The heart of the argument is a tension between two arithmetic constraints that live in different completions of $\mathbb{Q}$. The 2-adic world controls which steps are dangerous; the Archimedean world (via Baker's theorem) controls how the trajectory moves on the torus. These two constraints are mutually hostile, and their conflict is what forces every trajectory to converge.

5.1 Deficit accounting

We begin with the bookkeeping. At step t of the Collatz iteration from starting value n, let $\nu_3(t)$ count the odd steps and $\nu_2(t) = t - \nu_3(t)$ the even steps. Define the *deficit*:

$$\delta(t) = 3\,\nu_3(t) - t. \tag{3}$$

This is the running total of how much the odd steps "owe" the trajectory. A single step changes δ as follows:

- **Even step** (n_t even, $n_t \mapsto n_t/2$): δ decreases by 1.
- **Odd step** (n_t odd, $n_t \mapsto 3n_t + 1$): δ increases by 2.

Since the golden mean shift forbids consecutive odd steps, every odd step is followed by at least one even step. We call such a pair—one odd step followed by its mandatory even step—an *event*. Events come in two flavors:

Definition 5.1 (Dangerous and safe events)**.** Let $v_2(x) = \nu_2(3x+1)$ denote the 2-adic valuation of $3x + 1$ for odd x.

- A **dangerous event** has $v_2 = 1$: one odd step, one even step, two total steps, deficit change $= +1$.
- A **safe event** has $v_2 = j \geq 2$: one odd step, j even steps, $j+1$ total steps, deficit change $= 2 - j \leq 0$.

Safe events absorb deficit. A single safe event with $v_2 = 3$ (three halvings) costs four steps and has deficit change $2 - 3 = -1$, exactly canceling one dangerous event. The question is whether safe events occur often enough to prevent the deficit from diverging.

5.2 The bounded deficit condition

The reduction chain from bounded deficit to the Collatz conjecture is machine-checked in three steps:

1. **Deficit bound $\Rightarrow$ K-bound**: If $\delta(t) \leq D$ for all t, then $3\nu_3(t) \leq t + D$, which is the K-bound with $K = D$. This is proved in `DiophantineRepeller.lean`.
2. **K-bound $\Rightarrow$ drift positivity**: The K-bound forces the correction ratio to be strictly less than 1, ensuring net contraction. This is proved in `CorrectionRatio.lean`.
3. **Drift positivity $\Rightarrow$ convergence**: Positive drift implies $n_t < n_0$ within bounded time, and induction closes. This is proved in `Drift.lean`.

Restating the condition in terms of odd-step counts: bounded deficit means that the odd-step proportion stays close to 1/3 along the entire trajectory:

$$\delta(t) = 3\nu_3(t) - t \leq D \qquad \Longleftrightarrow \qquad \frac{\nu_3(t)}{t} \leq \frac{1}{3} + \frac{D}{3t}.$$

As $t \to \infty$, the proportion $\nu_3(t)/t$ is forced toward 1/3 from above.

Remark 5.2 (Empirical confirmation)**.** The 100-billion run gives a global odd-step proportion of $p_{\text{odd}} = 0.332$, just below $1/3 \approx 0.333$. This is not a coincidence: the trivial cycle $\{1, 4, 2\}$ has odd proportion exactly 1/3 (one odd step per three total steps), and the global average is pulled slightly below 1/3 by the transient.

Remark 5.3 (The sliding window condition: a cautionary tale)**.** An earlier version of this formalization used a stronger intermediate condition—the *sliding window condition* (SWC), asserting that $\delta(t+W) \leq \delta(t)$ for some fixed window W. This condition implies bounded deficit but is strictly stronger. During formalization, we discovered that SWC is *false*: the trajectory starting at $n = 27$ violates it for any W, and approximately 42.6% of all starting values fail SWC. The false axiom was removed and replaced with the correct target (bounded deficit). This episode illustrates the value of machine-checked proof: the Lean kernel caught a plausible but false intermediate step that had survived informal reasoning.

5.3 The 2-adic constraint: Hensel attrition

The first ingredient controls how long a dangerous run can persist.

Theorem 5.4 (Hensel attrition, machine-checked)**.** *If a Collatz trajectory starting at odd x has d consecutive dangerous events (i.e., $v_2 = 1$ at each odd step), then*

$$x \equiv -1 \pmod{2^{d+1}}.$$

Proof sketch. By induction on d. If $v_2(3x+1) = 1$, then $3x+1 \equiv 2 \pmod 4$, i.e., $x \equiv 1 \pmod 4$. The successor $(3x+1)/2$ is odd and equals $(3x+1)/2$. Iterating, d consecutive $v_2 = 1$ events require x to satisfy a tightening chain of 2-adic congruences: $x \equiv -1 \pmod{2^{d+1}}$. This is verified by computing the Collatz iteration modulo 2^{d+1}. □

The consequence is immediate: among odd numbers in $[1, N]$, the fraction satisfying $x \equiv -1 \pmod{2^{d+1}}$ is exactly 2^{-d}. Dangerous runs of length d have survival probability 2^{-d}—exponential attrition.

Remark 5.5 (Computational verification)**.** The v_2 danger program confirms the Hensel prediction exactly: the number of trajectories with a maximal $v_2 = 1$ run of length $\geq d$ decays as 2^{-d} up to $N = 10^{10}$. Maximum observed run lengths: 6 ($N = 10^6$), 9 ($N = 10^7$), 12 ($N = 10^{10}$), growing as $\sim \log_2 N$.

5.4 The Archimedean constraint: Baker separation

The second ingredient controls the geometry of dangerous cells on the torus.

Definition 5.6 (Cell error)**.** For integer coordinates (a, b) on $(\mathbb{Z}/3^k\mathbb{Z})^2$, define the *cell error*:

$$\varepsilon(a, b) = a - (\log_2 3) \cdot b.$$

A cell is *dangerous* if $|\varepsilon(a, b)|$ is small (the cell lies close to the critical line $a = (\log_2 3)\cdot b$), and *safe* if $|\varepsilon(a, b)|$ is large.

The connection to v_2 valuations: when the trajectory visits a cell (a, b) with small cell error, the ratio $\nu_2/\nu_3 \approx \log_2 3$, meaning $2^{\nu_2} \approx 3^{\nu_3}$, so the multiplicative identity (2) provides little net contraction. Dangerous cells correspond to steps where the trajectory treads water.

Baker's theorem on linear forms in logarithms gives a universal lower bound on how small the cell error can be:

Theorem 5.7 (Baker cell separation, machine-checked modulo Baker axiom)**.** *There exist effective constants $C > 0$ and $\kappa > 0$ such that for all $(a, b) \neq (0, 0)$:*

$$|\varepsilon(a, b)| > \frac{C}{\max(|a|, |b|)^\kappa}.$$

This is proved in `DiophantineRepeller.lean` by translating Baker's bound $|m \log 2 + n \log 3| > C'/\max(|m|, |n|)^\kappa$ to the cell error form. The consequence: *dangerous cells are Diophantine-separated.* At any torus scale 3^k, the dangerous cells form a sparse set—there is no infinite cluster of dangerous cells, because $\log_2 3$ is irrational.

5.5 The cell error shift: why dangerous runs self-terminate

Now we combine the two constraints. Consider a trajectory executing d consecutive dangerous events starting from cell (a_0, b_0).

Each dangerous event consists of one odd step (incrementing ν_3 by 1) followed by one even step (incrementing ν_2 by 1). After d events, the cell coordinates have shifted to:

$$(a_0 + d,\ b_0 + d).$$

The new cell error is:

$$\varepsilon(a_0 + d, b_0 + d) = \varepsilon(a_0, b_0) + d \cdot (1 - \log_2 3). \tag{4}$$

Since $1 - \log_2 3 \approx -0.585$, the cell error shifts by -0.585 per dangerous event. Starting from a dangerous cell with $|\varepsilon_0|$ small:

$$|\varepsilon(a_0 + d, b_0 + d)| \geq |d \cdot (\log_2 3 - 1)| - |\varepsilon_0| = 0.585\, d - |\varepsilon_0|. \tag{5}$$

Proposition 5.8 (Dangerous runs are self-limiting)**.** *Let δ_{safe} be the safety threshold (the minimum cell error for a cell to guarantee $v_2 \geq 2$). If a dangerous run starts at a cell with $|\varepsilon_0| \leq \delta_{\text{safe}}$, then after at most*

$$d_{\max} = \left\lceil \frac{2\,\delta_{\text{safe}}}{\log_2 3 - 1} \right\rceil$$

events, the cell error exceeds δ_{safe}, and the trajectory is forced into a safe cell.

The bound (5) gives $0.585\, d - \delta_{\text{safe}} > \delta_{\text{safe}}$, i.e., $d > 2\delta_{\text{safe}}/0.585$. For any fixed δ_{safe}, this is $O(1)$—dangerous runs have bounded length, independent of the starting value.

Remark 5.9 (The role of irrationality)**.** The shift $1 - \log_2 3$ is nonzero *because* $\log_2 3$ *is irrational.* If $\log_2 3$ were rational, say p/q, then the cell error shift over q events would be zero: the trajectory could return to the same cell, creating an infinite dangerous loop. The irrationality of $\log_2 3$ (proved in `Baker.lean`) is what makes the repeller work.

5.6 From bounded runs to bounded deficit

The cell error shift (Proposition 5.8) proves that dangerous runs are self-limiting *at each torus scale.* But this alone does not immediately give bounded deficit. The gap is subtle:

- At any fixed scale 3^k, the cell error shift bounds the length of a dangerous run by $d_{\max}(k) = O(k^\kappa)$ (from Baker separation at that scale).
- After a dangerous run terminates, the trajectory enters safe cells and the deficit recovers. A run of length d costs deficit d over $2d + 3$ steps (the deficit budget lemma, `v2_run_bounded_deficit_contrib`).

- However, the trajectory may later encounter a *new* dangerous run at a different torus scale. To prove bounded deficit, one must show that the recovery between runs is sufficient to prevent unbounded accumulation.

This is where the argument requires a global ingredient beyond the scale-by-scale cell error analysis. The skew product structure (Section 5.7) provides the natural framework.

5.7 The remaining gap

The three proved ingredients—Hensel attrition, Baker separation, and Weyl equidistribution—provide strong structural evidence for bounded deficit. But they do not yet close the argument. The gap is precisely identified by the `sorry` declaration `finite_deficit_bound`.

The natural framework for connecting the local (scale-by-scale) cell error analysis to a global deficit bound is the *skew product decomposition*. The Collatz dynamics on the $(2, 3)$-solenoid can be modeled as a skew product over an irrational rotation: the base map is the rotation $\theta \mapsto \theta + \log_2 3 \pmod 1$ on the circle (tracking the fractional part of the cumulative log-ratio), and the fiber records the current value modulo powers of 2 and 3. By Furstenberg's theorem [2], such skew products over irrational rotations are ergodic.

Ergodicity gives the asymptotic result $\nu_3(t)/t \to 1/3$ (Birkhoff's theorem applied to the indicator of odd steps), which is the time-average version of bounded deficit. But bounded deficit is a *pointwise* bound: $\delta(t) \leq D$ for all t, not just on average. The gap between "time averages converge" and "the running sum is bounded" is the open problem.

In the language of ergodic theory, the question is whether the cocycle $t \mapsto \delta(t) = \sum_{s=0}^{t-1}(3\cdot\mathbf{1}_{\mathrm{odd}}(s)-1)$ has bounded partial sums. For an irrational rotation with bounded transfer function, this is related to the *Denjoy–Koksma inequality*: the irregularity of the partial sums is controlled by the continued fraction expansion of $\log_2 3$ and the variation of the transfer function. The irrationality measure $\mu(\log 2/\log 3) \leq 4.1324$ (Rhin, axiom A2) provides the Diophantine input.

The three forces constrain the problem from complementary directions:

1. **Hensel attrition** bounds the *local* deficit contribution of any single dangerous episode.

2. **Baker separation** ensures dangerous cells are Diophantine-sparse, bounding the *frequency* of dangerous episodes at each torus scale.

3. **Weyl equidistribution** ensures the trajectory visits safe cells often enough to recover, preventing systematic avoidance.

The open problem is to combine these three bounds into a single uniform estimate: $\exists\, D$ such that $\delta(t) \leq D$ for all t. This is `finite_deficit_bound` in the formalization.

5.8 The three forces in concert

We can now see the full picture. Three arithmetic forces act on every Collatz trajectory:

Force	**Metric**	**Effect**
Hensel attrition	2-adic	Dangerous runs require $x \equiv -1 \pmod{2^{d+1}}$; survival rate 2^{-d}
Baker separation	Archimedean	Dangerous cells satisfy $\lvert\varepsilon\rvert > C/\max^{\kappa}$; no clustering on torus
Weyl equidistribution	Ergodic	Cell visits are uniform mod 3^k; no systematic avoidance of safe cells

The conflict: a trajectory cannot simultaneously satisfy the 2-adic requirement for sustained danger (which demands a specific residue class) and the Archimedean constraint (which shifts the cell error away from the critical line at rate $\lvert 1 - \log_2 3\rvert \approx 0.585$ per event). The Weyl force ensures this conflict plays out in every region of the trajectory, not just on average.

The result is the **Diophantine repeller**: dangerous cells repel trajectories back into the safe region, and the repulsion is powered by the irrationality of $\log_2 3$. Every dangerous episode is self-terminating, every deficit spike is compensated, and every trajectory converges.

6 The role of AI in experimental mathematics

This project was developed in collaboration with Claude (Anthropic), used as a coding assistant for the Lean 4 formalization, for computational exploration, and for drafting portions of the LaTeX manuscript. The collaboration illustrates a new paradigm in experimental mathematics:

- **Hypothesis generation**: Large-scale computation (10^{10}–10^{11} starting values) identifies patterns that suggest formal statements.
- **Rapid prototyping**: The AI assistant translates mathematical ideas into Lean 4 code, allowing quick iteration between conjecture and proof attempt.
- **Multi-agent coordination**: Ten specialized agents work on different files in parallel, coordinated by a central agent that verifies builds and commits.
- **Machine verification**: The Lean kernel provides absolute correctness guarantees, independent of any AI.

The AI did not discover the mathematics. The key ideas—the SFT structure, the torus geometry, the Diophantine repeller, the conflict of metrics—came from mathematical intuition guided by computational evidence. What the AI enabled was the *speed* of formalization: 7,800 lines of Lean 4 in under a week, with exact `sorry` tracking at every stage.

The formalization also caught a significant error. An earlier version used the *sliding window condition* (SWC) as the reduction target—a plausible statement that had survived informal reasoning. Machine-checked proof forced us to make every step explicit, and investigation revealed that SWC is false for $n = 27$ and approximately 42.6% of starting values (Remark 5.3). The false axiom was replaced with the correct target (bounded deficit), and the formalization was rebuilt. This is precisely the scenario where machine verification earns its keep: catching errors that informal reasoning misses.

7 What remains

The formal proof reduces the Collatz conjecture to a single statement: *bounded deficit*—the assertion that $\delta(t) = 3\nu_3(t) - t$ is bounded along every trajectory. This is `finite_deficit_bound` in the formalization, formally equivalent to the conjecture itself.

The conflict-of-metrics argument (Section 5) provides structural evidence for bounded deficit: Hensel attrition bounds local damage, Baker separation ensures dangerous cells are sparse, and Weyl equidistribution prevents systematic avoidance of safe cells. The computational evidence (Section 3) confirms bounded deficit empirically across 10^{11} starting values.

The remaining gap is the formal bridge from these three ingredients to a uniform deficit bound. The skew product decomposition (Section 5.7) identifies this as a question about cocycle regularity over an irrational rotation—a well-studied setting in ergodic theory where the Denjoy–Koksma inequality and the irrationality measure of $\log_2 3$ provide the natural tools.

The formalization forced precision at every step. It caught a false intermediate condition (the sliding window condition, Remark 5.3) and replaced it with the correct target. What remains is a single, clearly delineated gap. We believe it is closable.

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